

Khwaja Yunus Ali University

School of Science and Engineering

B.Sc. in CSE & EEE

Lesson Plan

Course Title : Physics - I

Course Code : CSE 0533 1101 (CSE) & PHY 0533 1101 (EEE)

Term: Spring – 2024	Date: 22.01.2024	Time: 10.00 AM	Room: 327	Activity: Lecture 7
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Topics to be discussed: Electric flux, Gauss' law, Applications of it.

Target CLO & PLO : CLO2-3 & PLO 2 & 5

Learning Outcomes : At the end of the class the students will be able to:

- ❖ Define electric flux and charge density
- ❖ State and explain Gauss's Law
- ❖ Explain how to derive Coulomb's Law from Gauss's Law
- ❖ Applications of Gauss's Law

Methods of Teaching:

Mode/Activities	Time
Feedback of previous class	10min
Interactive discussion	10min
Lecture Presentation on Topics	30 min
Group activity	25 min
Moral & Ethical awareness & Feedback session	10 min

Assessment Strategy : Quizzes/Class tests, Class Participation

Sample Questions (MCQ):

- 1) $\oint_s \vec{E} \cdot d\vec{a} = ?$
a) Coulomb b) Potential c) Energy d) Force e) Electric flux
- 2) Electric flux equal to time of the total charge enclosed by the surface.
a) $1/\pi$ b) $1/\mu$ c) $1/\epsilon_0$ d) ϵ_0 e) π

Sample Questions (SAQ):

- 1) Define electric flux & charge density.
- 2) Explain the Gaussian surface.
- 3) State and explain Gauss's Law.
- 4) Deduce Coulomb's law from Gauss's law.
- 5) Using Gauss's law, calculate the electric field at a distance y from an infinite line of charge of with uniform linear charge density λ .
- 6) Prove that the net flux of a cylindrical Gaussian surface placed in an external electric field is zero.
- 7) An electric field of 500V/m makes an angle 60° with the surface vector which has a magnitude of 0.500m^2 . Find the flux that passes through the surface.
- 8) A point charge of $2.0\mu\text{C}$ is at the center of a cubic Gaussian surface 8.0cm edge. What the net electric flux through the surface?

Reference Books:

1. Giasuddin ahmed – Physics for Engineers (Part-2)
2. Huq. Rafiqullah- Electricity and Magnetism

Pre-reading book for next Lecture:

- Md. Ziaul Ahsan – Physics for Engineering (Lecture Series), Pages: 332-342
- Dr. Amir Hussain Khan – Physics 2nd Paper (HSC Text Book), Pages: 37-86